

Gas Optimization Report: RWA T-Bill Tokenization Platform

Scope: Gas-sensitive protocol operations and required L1 vs L2 comparison table

Tooling: Foundry gas snapshots through `script/GasCompare.s.sol`

Local chain: Anvil chain ID 31337

L2 target: Base Sepolia

Status: Local gas measured. L1 and Base Sepolia columns are filled after real RPC deployment secrets are provided.

1. Executive Summary

The protocol keeps gas-sensitive logic concentrated in four areas: asset minting, ERC4626 vault accounting, AMM liquidity and swap math, and oracle reads. The largest local cost is AMM liquidity addition because it includes two token transfers, reserve updates, LP share minting, and first-liquidity square-root calculation.

17,308

Local gas for `swapExactIn` in the benchmark path.

2,736

Local gas for adapter `latestPrice` validation.

152,599

Local gas for full AMM add-liquidity path.

2. Measurement Method

The gas numbers are produced by `script/GasCompare.s.sol` against a deployed local protocol. The script performs representative operations rather than isolated opcode microbenchmarks, so each number includes realistic protocol overhead such as role checks, ERC20 accounting, SafeERC20 calls, reserve synchronization, and event emission where applicable.

Recommended commands:

```
npm run anvil  
npm run deploy:anvil  
forge script script/GasCompare.s.sol:GasCompare --rpc-url http://127.0.0.1:8545
```

For real comparison, run the same script against Ethereum Sepolia or a mainnet fork for L1-style gas and against Base Sepolia for L2 deployment gas.

3. Required L1 vs L2 Comparison Table

Operation	Local Anvil gas	L1 measured	Base Sepolia measured	Notes
TBill mint	26,003	TBD	TBD	Role-gated ERC20 mint by authorized issuer.
Vault deposit	79,670	TBD	TBD	ERC4626 share mint plus SafeERC20 transfer.
Vault reportYield	7,519	TBD	TBD	Role-gated yield transfer into the vault.
AMM addLiquidity	152,599	TBD	TBD	Two token transfers, LP minting, reserve sync, and initial square-root calculation.
AMM swapExactIn	17,308	TBD	TBD	Fee math, slippage check, output transfer, and invariant check.
Oracle latestPrice	2,736	TBD	TBD	Feed read plus stale, zero, and negative answer validation.
Treasury depositToken	28,994	TBD	TBD	Pull-payment accounting and SafeERC20 transfer.

4. Optimization Decisions

Decision	Before / alternative	After / implementation	Reason
Inline Yul square root	Pure Solidity Babylonian helper	<code>RwaMath.sqrtYul</code> for AMM initial liquidity	Satisfies assembly requirement and keeps the hot calculation compact.
Pull-payment treasury	Push transfer during scheduling	Schedule owed balance, payee withdraws later	Avoids failed payout loops and makes external call surface smaller.
Non-upgradeable supporting contracts	Proxy every contract	Only <code>TBillToken</code> is UUPS upgradeable	Reduces deploy and runtime complexity for AMM, vault, oracle, treasury, and governor.
Immutable AMM token addresses	Storage reads for token addresses	Constructor immutables	Avoids repeated storage reads in swap and liquidity paths.
Adapter-level oracle validation	Duplicate validation in each consumer	Centralized <code>OracleAdapter</code> checks	Keeps feed safety logic reusable and auditable.

5. Yul vs Solidity Benchmark Context

`RwaMath.sol` contains both `sqrtSolidity` and `sqrtYul`. The production AMM path uses the Yul helper only during first-liquidity LP minting. Keeping the assembly block small is intentional: it demonstrates the advanced Solidity requirement without hiding broad protocol logic inside assembly.

Security note: the Yul helper is deterministic, does not perform external calls, and is covered by tests comparing behavior against the pure Solidity equivalent.

6. Final Submission Steps

Before final submission, fill the L1 and Base Sepolia columns by running the same benchmark script against real RPC endpoints:

```
forge script script/GasCompare.s.sol:GasCompare --rpc-url "$SEPOLIA_RPC_URL"  
forge script script/GasCompare.s.sol:GasCompare --rpc-url "$BASE_SEPOLIA_RPC_URL"
```

Then update this report with transaction hashes or script output references. The current local report is complete for Anvil and ready to be extended once deployment secrets are available.